

Hermitian line bundles

Let K be a number field with ring of integers $\mathcal{O}_K$. Let $\mathcal{X}$ be an arithmetic variety over $\mathrm{Spec} \mathcal{O}_K$ of relative dimension n . **Yang:** flat, separated, of finite type,

For every $\sigma : \mathcal{O}_K \rightarrow \mathbb{C}$, we denote by $\mathcal{X}_\sigma(\mathbb{C}) := \mathcal{X} \times_{\mathrm{Spec} \mathcal{O}_K, \sigma} \mathrm{Spec} \mathbb{C}$ the complex analytic space associated to the base change of $\mathcal{X}$ along σ .

1 Definitions and examples

Definition 1. A hermitian line bundle $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$ on $\mathcal{X}$ consists of

- a line bundle $\mathcal{L}$ on $\mathcal{X}$;
- for each embedding $\sigma : K \hookrightarrow \mathbb{C}$ (equivalently, each infinite place $\sigma \in M_K^\infty$), a smooth hermitian metric $\|\cdot\|_\sigma$ on the complex line bundle $\mathcal{L}_\sigma(\mathbb{C})$ over $\mathcal{X}_\sigma(\mathbb{C})$, which is invariant under the complex conjugation map.

Yang: To be checked.

Example 2. Consider $\mathcal{X} = \mathbb{P}_{\mathcal{O}_K}^n$ and $\mathcal{L} = \mathcal{O}_{\mathbb{P}_{\mathcal{O}_K}^n}^n(1)$. There is a class of natural hermitian metrics on $\mathcal{O}_{\mathbb{P}_{\mathcal{O}_K}^n}^n(1)_\sigma(\mathbb{C})$ called the *Fubini-Study metrics* constructed as follows.

Recall that $\mathbb{P}^n$ represents the functor that sends a scheme S to the set of isomorphism classes of pairs $(\mathcal{E}, \phi)$, where $\mathcal{E}$ is a rank $n+1$ vector bundle on S and $\phi : \mathcal{E} \rightarrow \mathcal{L}$ is a surjective morphism onto a line bundle $\mathcal{L}$ on S . And the universal pair on $\mathbb{P}^n$ is given by $\mathcal{O}_{\mathbb{P}^n}^{\oplus n+1} \rightarrow \mathcal{O}_{\mathbb{P}^n}(1)$.

For each $\sigma \in M_K^\infty$, fix a hermitian metric $\|\cdot\|$ on $\mathbb{C}^{n+1}$. Every point $x \in \mathbb{P}_{\mathcal{O}_K}^n(\sigma)$, it corresponds to a surjective morphism $\phi_x : \mathbb{C}^{n+1} \rightarrow L_x$ with L_x the fiber of $\mathcal{L}_\sigma(\mathbb{C})$ at x . Explicitly, every point $x = [z_0 : z_1 : \cdots : z_n] \in \mathbb{P}^n(\mathbb{C})$ corresponds to the surjective morphism

$$\phi_{[z_0:z_1:\dots:z_n]} : \mathbb{C}^{n+1} \rightarrow \mathbb{C} \cong L_x, \quad (x_0, x_1, \dots, x_n) \mapsto x_0 z_0 + x_1 z_1 + \cdots + x_n z_n.$$

The fixed hermitian metric induces the quotient hermitian metric $\|\cdot\|_x$ on L_x . Hence we obtain a hermitian metric $\|\cdot\|_\sigma$ on $\mathcal{O}_{\mathbb{P}_{\mathcal{O}_K}^n}^n(1)_\sigma(\mathbb{C})$. This construction defines a hermitian line bundle $\overline{\mathcal{O}_{\mathbb{P}_{\mathcal{O}_K}^n}^n(1)}$.

Yang: To be checked.

Definition 3. Let $\overline{\mathcal{L}}_1$ and $\overline{\mathcal{L}}_2$ be hermitian line bundles on $\mathcal{X}$.

- The *tensor product* $\overline{\mathcal{L}}_1 \otimes \overline{\mathcal{L}}_2$ is defined as the line bundle $\mathcal{L}_1 \otimes \mathcal{L}_2$ on $\mathcal{X}$ equipped with the metrics $\|s_1 \otimes s_2\|_\sigma = \|s_1\|_{\sigma,1} \cdot \|s_2\|_{\sigma,2}$ on $(\mathcal{L}_1 \otimes \mathcal{L}_2)_\sigma(\mathbb{C})$
- The *dual* $\overline{\mathcal{L}}_1^\vee$ is defined as the dual line bundle $\mathcal{L}_1^\vee$ on $\mathcal{X}$ equipped with the dual metrics $\|\cdot\|_{\sigma,1}^\vee$ on $(\mathcal{L}_1^\vee)_\sigma(\mathbb{C})$ for each $\sigma \in M_K^\infty$.

Yang: To be checked.

Proposition 4. The set of isomorphism classes of hermitian line bundles on $\mathcal{X}$ forms an abelian group under the tensor product operation. The group is called the *arithmetic Picard group* of $\mathcal{X}$ and is denoted by $\widehat{\text{Pic}}(\mathcal{X})$.

Proof. Direct verification. $\square$

Note that for every $\sigma \in M_K^\infty$, any two smooth hermitian metrics on $\mathcal{L}_\sigma(\mathbb{C})$ differ by a smooth function on $\mathcal{X}_\sigma(\mathbb{C})$. Explicitly, for any two smooth hermitian metrics $\|\cdot\|_\sigma$ and $\|\cdot\|'_\sigma$ on $\mathcal{L}_\sigma(\mathbb{C})$, there exists a unique smooth function $f : \mathcal{X}_\sigma(\mathbb{C}) \rightarrow \mathbb{R}$ such that

$$\|s\|'_\sigma = e^{-f} \|s\|_\sigma$$

for all local sections s of $\mathcal{L}_\sigma(\mathbb{C})$. Conversely, given any smooth hermitian metric $\|\cdot\|_\sigma$ on $\mathcal{L}_\sigma(\mathbb{C})$ and any smooth function $f : \mathcal{X}_\sigma(\mathbb{C}) \rightarrow \mathbb{R}$, we can define a new smooth hermitian metric $\|\cdot\|'_\sigma$ on $\mathcal{L}_\sigma(\mathbb{C})$ by the above formula.

Example 5. Let $\mathcal{X} = \mathbb{P}_\mathbb{Z}^n$. Then we have

$$\widehat{\text{Pic}}(\mathbb{P}_\mathbb{Z}^n) \cong \mathbb{Z} \times \mathcal{C}^\infty(\mathbb{P}^n(\mathbb{C}), \mathbb{R}).$$

Here the generator of the first factor $\mathbb{Z}$ corresponds to the hermitian line bundle $\overline{\mathcal{O}_{\mathbb{P}_\mathbb{Z}^n}(1)}$ constructed in [Example 2](#).

Definition 6. Let $f : \mathcal{Y} \rightarrow \mathcal{X}$ be a morphism of arithmetic varieties over $\text{Spec } \mathcal{O}_K$. Let $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$ be a hermitian line bundle on $\mathcal{X}$. The *pullback* $f^*\overline{\mathcal{L}}$ is defined as the line bundle $f^*\mathcal{L}$ on $\mathcal{Y}$ equipped with the metrics $\|s\|'_\sigma = \|f_\sigma^* s\|_\sigma$ on $(f^*\mathcal{L})_\sigma(\mathbb{C})$ for each $\sigma \in M_K^\infty$, where $f_\sigma : \mathcal{Y}_\sigma(\mathbb{C}) \rightarrow \mathcal{X}_\sigma(\mathbb{C})$ is the induced morphism of complex manifolds. **Yang: To be checked.**

Definition 7. Let $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$ be a hermitian line bundle on $\mathcal{X}$. The set of *small sections* of $\overline{\mathcal{L}}$ is defined as

$$H^0(\mathcal{X}, \overline{\mathcal{L}}) := \{s \in H^0(\mathcal{X}, \mathcal{L}) : \|s(x)\|_\sigma \leq 1, \forall \sigma \in M_K^\infty, x \in \mathcal{X}_\sigma(\mathbb{C})\}.$$

Yang: To be checked.

Example 8. Let $\mathcal{X} = \mathbb{P}_\mathbb{Z}^n$ and $\overline{\mathcal{L}} = \overline{\mathcal{O}_{\mathbb{P}_\mathbb{Z}^n}(1)}$ be the hermitian line bundle constructed in [Example 2](#). We have

$$H^0(\mathbb{P}_\mathbb{Z}^n, \mathcal{L}) = \mathbb{Z}X_0 \oplus \mathbb{Z}X_1 \oplus \cdots \oplus \mathbb{Z}X_n,$$

where $X_0, X_1, \dots, X_n$ are the standard homogeneous coordinates on $\mathbb{P}_\mathbb{Z}^n$. For every $s = \sum a_n X_n \in H^0(\mathbb{P}_\mathbb{Z}^n, \mathcal{L})$ and $x = [x_0 : \cdots : x_n] \in \mathbb{P}^n(\mathbb{C})$, we have

$$\|s(x)\| = \inf_{x_0 y_0 + \cdots + x_n y_n = 0} \left(\sum |a_k + y_k|^2 \right)^{1/2} = \frac{|\sum a_k x_k|}{\|x\|},$$

where $\|x\| = (|x_0|^2 + |x_1|^2 + \cdots + |x_n|^2)^{1/2}$. When x varies, by Cauchy-Schwarz inequality, we have

$$\sup_{x \in \mathbb{P}^n(\mathbb{C})} \|s(x)\| = \sup_{x \in \mathbb{P}^n(\mathbb{C})} \frac{|\sum a_k x_k|}{\|x\|} = \|a\|.$$

Note that $a_n \in \mathbb{Z}$ for all n . Therefore, we have

$$H^0(\mathbb{P}_\mathbb{Z}^n, \overline{\mathcal{O}_{\mathbb{P}_\mathbb{Z}^n}(1)}) = \{0, \pm X_0, \pm X_1, \dots, \pm X_n\}$$

and

$$\#H^0(\mathbb{P}_{\mathbb{Z}}^n, \overline{\mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(1)}) = 2n + 3.$$

Proposition 9. The set $H^0(\mathcal{X}, \overline{\mathcal{L}})$ is finite.

Definition 10. Let $\overline{\mathcal{L}}$ be a hermitian line bundle on $\mathcal{X}$. We set

$$h^0(\mathcal{X}, \overline{\mathcal{L}}) := \log \#H^0(\mathcal{X}, \overline{\mathcal{L}}).$$

The *volume* of $\overline{\mathcal{L}}$ is defined as

$$\text{vol}(\overline{\mathcal{L}}) := \limsup_{m \rightarrow \infty} \frac{h^0(\mathcal{X}, \overline{\mathcal{L}}^{\otimes m})}{m^{n+1}/(n+1)!}.$$

Example 11. We continue with [Example 8](#). For every integer $m \geq 1$, we have

$$H^0(\mathbb{P}_{\mathbb{Z}}^n, \mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(m)) = \bigoplus_{|\alpha|=m} \mathbb{Z}X^\alpha.$$

Let $t = \sum_{|\alpha|=m} a_\alpha X^\alpha \in H^0(\mathbb{P}_{\mathbb{Z}}^n, \mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(m))$ be a global section. Denote by M (resp. L) the geometric vector bundle associated to $\mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(m)$ (resp. $\mathcal{O}_{\mathbb{P}_{\mathbb{Z}}^n}(1)$).

We first focus on the affine chart $U_0 = \{X_0 \neq 0\}$ and let $s = X_0 \in H^0(\mathbb{P}^n, \mathcal{O}(1))$. For every point $x = [1 : x_1 : \dots : x_n] \in U_0(\mathbb{C})$,

$$\|s(x)\| = \frac{1}{(1 + |x_1|^2 + \dots + |x_n|^2)^{1/2}} \implies \|t(x)\| = \frac{|\sum a_\alpha x^\alpha|}{(1 + |x_1|^2 + \dots + |x_n|^2)^{m/2}}.$$

Hence one can see that for any point $x = [x_0 : \dots : x_n] \in \mathbb{P}^n(\mathbb{C})$, we have

$$\|t(x)\| = \frac{|\sum a_\alpha x^\alpha|}{\|x\|^m}$$

with $\|x\| = (|x_0|^2 + |x_1|^2 + \dots + |x_n|^2)^{1/2}$.

Yang: In general, it seems difficult to compute $h^0(\mathbb{P}_{\mathbb{Z}}^n, \overline{\mathcal{O}(m)})$ explicitly. For example, when $m = 2$, we have that $\sup_{x \in \mathbb{P}^n(\mathbb{C})} \|t(x)\|$ is equal to the largest norm of eigenvalues of the matrix associated to the quadratic form (a_{ij}) . Hence we just compute its asymptotic behavior when $m \rightarrow \infty$. Note that we have

$$\frac{|\sum x^\alpha|}{\|x\|^m} \cdot \min_{|\alpha|=m} |a_\alpha| \leq \|t(x)\| \leq \frac{|\sum x^\alpha|}{\|x\|^m} \cdot \max_{|\alpha|=m} |a_\alpha|.$$

By the lore of inequalities (like Karamata's inequality), one can show that

$$\sup_{x \in \mathbb{P}^n(\mathbb{C})} \frac{|\sum x^\alpha|}{\|x\|^m} = \binom{m+n}{n} \frac{1}{(n+1)^{m/2}}.$$

Yang: To be checked.

2 Arithmetic intersection theory

Appendix

DRAFT